

Another Geometric Interpretation of Cramer's Rule

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In introductory linear algebra, matrix-vector multiplication $A\mathbf{x} = \mathbf{b}$ is often characterized by treating the elements of $\mathbf{b}$ as the inner product of the row vectors of A with $\mathbf{x}$. In the author's experience, it is less frequently but more intuitively characterized by treating $\mathbf{b}$ as the linear combination of the column vectors of A , as in $\mathbf{b} = \mathbf{a}_1x_1 + \cdots + \mathbf{a}_nx_n$. When $\mathbf{b}$ is given and $\mathbf{x}$ is unknown, these characterizations lead to different visualizations of the solution to systems of linear equations. We will illustrate with a non-singular example in $\mathbb{R}^3$, with

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 1 & -1 & 4 \\ 4 & 1 & -1 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}. \quad (0.1)$$

To visualize the problem treating the multiplication as the inner product of the row vectors of A with $\mathbf{x}$, we can consider the hyper-planes satisfying each row of the system of equations. The solution(s) to the system are the intersection of the hyper-planes (if they intersect). The three planes in the example are defined by the equations

$$\begin{aligned} 1 &= x_1 + 3x_2 + 2x_3, \\ 3 &= x_1 - x_2 + 4x_3, \text{ and} \\ 2 &= 4x_1 + x_2 - x_3 \end{aligned}$$

and illustrated in Figure 0.1. The hyper-planes are perpendicular to the rows of A giving some geometric meaning to this approach, but finding the perpendicular hyper-planes takes a step away from the original elements of A and $\mathbf{b}$ which may carry meaning. This geometric interpretation is even further from the solution methods associated with this characterization of matrix-vector multiplication, which tend to be algorithmic and numeric (e.g., Gaussian elimination.) The solution to the example problem is given by

$$\mathbf{x} = [20/29, -7/29, 15/29]^T.$$

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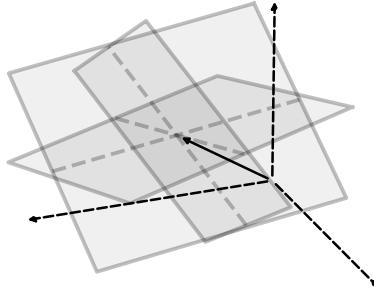


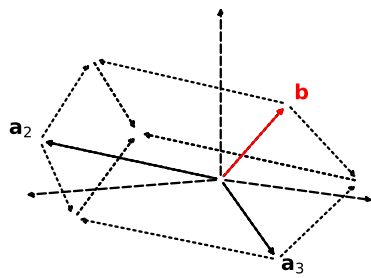
Figure 0.1: The hyper-plane representation of the solution to the example problem.

Alternatively, we can visualize the multiplication as a linear combination of the column vectors of A by interpreting the coefficients x_j as the application of Cramer's Rule. Cramer's Rule for non-singular A finds the j th coefficient as

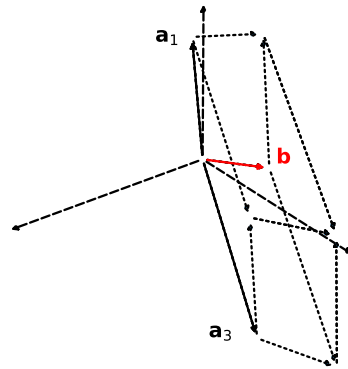
$$x_j = \frac{\det B_j}{\det A}, \quad (0.2)$$

where the matrix B_j is the matrix A with the j th column substituted with the vector $\mathbf{b}$. Orr presented a geometric approach to Cramer's rule in two or three dimensions by noting that the magnitudes of the determinants in (0.2) are equal to the areas of the parallelograms or volumes of parallelepipeds constructed from the columns of the B_j and A matrices [4]. In Figure 0.2, we construct parallelepipeds from the columns of the matrices used in Cramer's rule for Example (0.1). This volumetric interpretation has a direct connection to the components of the problem and adds meaning to the determinant operator.

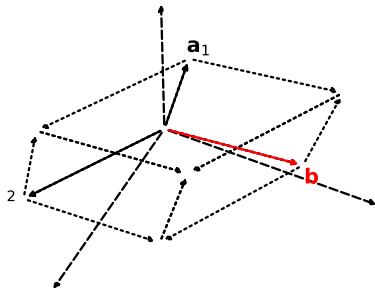
In this article, we develop another geometric interpretation of Cramer's rule by finding the coefficients of x_j as a generalization of projection onto orthogonal basis vectors using the rows of the adjugate. This interpretation makes more connections between elementary linear algebra concepts like the solution to linear equations, inner products, and projections. Such connections are useful for introducing broader concepts related to Hilbert spaces. The inner product feature of Hilbert spaces provide geometric interpretations beyond $\mathbb{R}^n$ like spectral and pseudospectral expansions in $\mathcal{L}^p$ function spaces. By comparing to the generalized cross product, this interpretation may also serve as an introduction to geometric algebras like Grassman algebra.



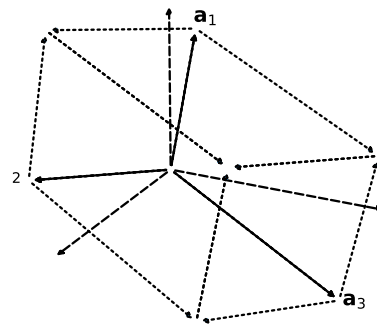
(a) Parallelepiped constructed from $\mathbf{b}$, $\mathbf{a}_2$, and $\mathbf{a}_3$



(b) Parallelepiped constructed from $\mathbf{a}_1$, $\mathbf{b}$, and $\mathbf{a}_3$



(c) Parallelepiped constructed from $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{b}$



(d) Parallelepiped constructed from $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$

Figure 0.2: Parallelepipeds constructed from the columns of the matrices used in Cramer's rule for the example problem.

Cramer's Rule

We'll re-write Cramer's rule (0.2) for a non-singular matrix A using the adjugate of A , $\text{adj } A = C^T$, which satisfies

$$AC^T = C^T A = (\det A) I. \quad (0.3)$$

The symbol C^T is used because the adjugate is equal to the transpose of the matrix of cofactors, C . Using (0.3), we can write two important statements. First, we can write the elements of the matrix multiplication $C^T A$ as

$$\mathbf{c}_i^T \mathbf{a}_j = \begin{cases} \det A & \text{if } i = j, \\ 0 & \text{if } i \neq j, \end{cases} \quad (0.4)$$

where $\mathbf{c}_i$ and $\mathbf{a}_j$ are the i th and j th columns of C and A , respectively. Second, we can find the inverse of A as $A^{-1} = \frac{1}{\det A} C^T$ to solve for $\mathbf{x}$, yielding

$$\mathbf{x} = \frac{1}{\det A} C^T \mathbf{b}.$$

If we compare the j th row of this matrix-vector multiplication to (0.2), we must have

$$\mathbf{c}_j^T \mathbf{b} = \det B_j. \quad (0.5)$$

Recognizing the left hand sides of equations (0.4) and (0.5) as inner products in $\mathbb{R}^n$, we can re-write Cramer's rule as

$$x_j = \frac{\langle \mathbf{c}_j, \mathbf{b} \rangle}{\langle \mathbf{c}_j, \mathbf{a}_j \rangle}. \quad (0.6)$$

We can use this version of Cramer's rule to write an expansion of $\mathbf{b}$ in terms of a non-orthogonal and non-normalized set of vectors $\mathbf{a}_j$,

$$\mathbf{b} = \sum_{j=1}^n \frac{\langle \mathbf{c}_j, \mathbf{b} \rangle}{\langle \mathbf{c}_j, \mathbf{a}_j \rangle} \mathbf{a}_j. \quad (0.7)$$

Considering Cramer's rule as a vectorial expansion can provide insights for and enhance connections between linear algebra concepts. We'll consider several in the next section.

Using the relationship

$$C^T = (\det A) A^{-1}, \quad (0.8)$$

we can find the adjugate for the example A as

$$C^T = \begin{bmatrix} -3 & 5 & 14 \\ 17 & -9 & -2 \\ 5 & 11 & -4 \end{bmatrix}. \quad (0.9)$$

Then we can compute the associated inner products as

$$\begin{aligned} \langle \mathbf{c}_1, \mathbf{b} \rangle &= 40, & \langle \mathbf{c}_2, \mathbf{b} \rangle &= -14, & \langle \mathbf{c}_3, \mathbf{b} \rangle &= 30, \\ \text{and } \langle \mathbf{c}_1, \mathbf{a}_1 \rangle &= \langle \mathbf{c}_2, \mathbf{a}_2 \rangle = \langle \mathbf{c}_3, \mathbf{a}_3 \rangle &= \det A &= 58. \end{aligned}$$

We illustrate the $\hat{\mathbf{c}}_i$ unit vectors parallel to the columns of C and the associated inner products in Figure 0.3.

Properties

Columns of the adjugate as a generalized cross product

Using the properties of the generalized cross product, such as described by Dittmer [2] or Shaw [5], we can recognize that $\mathbf{c}_j$ is related to the generalized cross product of the $n-1$ columns of A excluding $\mathbf{a}_j$. The equivalence between the columns of the matrix of cofactors and the generalized cross product was also recognized by Lightstone [3]. This generalized cross product is also equivalent to the wedge product of Grassman algebra [1], where these geometric relations are well described.

The handedness of this generalized cross product can be seen from the algebraic definition. Following Shaw, the generalized cross product of $n-1$ vectors satisfies

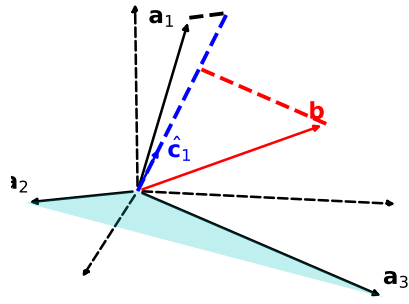
$$\det[\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{n-1}, \mathbf{a}_n] = (\mathbf{a}_1 \times \mathbf{a}_2 \times \dots \times \mathbf{a}_{n-1}) \cdot \mathbf{a}_n$$

for any $\mathbf{a}_n \in \mathbb{R}^n$. Since we have $\mathbf{c}_j \cdot \mathbf{a}_j = \det A$ in (0.4), due to the skew symmetry of the determinant operator, we relate the columns of the matrix of cofactors to the generalized cross product as

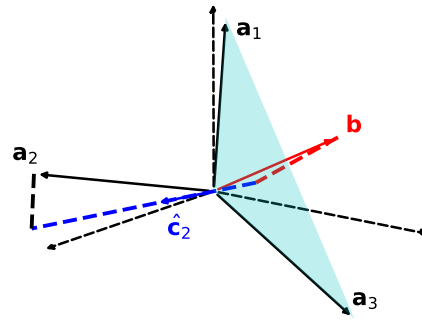
$$\mathbf{c}_j = (-1)^{n-j} (\mathbf{a}_1 \times \mathbf{a}_2 \times \dots \times \mathbf{a}_{j-1} \times \mathbf{a}_{j+1} \times \dots \times \mathbf{a}_{n-1} \times \mathbf{a}_n). \quad (0.10)$$

For the Example (0.1), we can confirm this relationship for the special case of three dimensions by observing that the $\mathbf{c}_j$ vectors, the rows of (0.9), are equal to the cross products of the remaining columns of A computed in a typical way with the change of sign as indicated by (0.10).

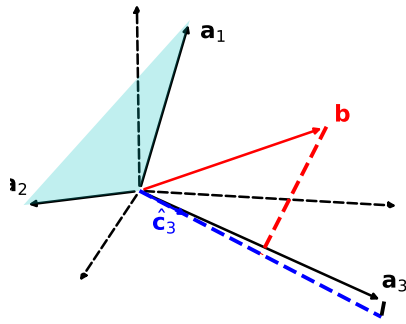
This interpretation is consistent with Orr's geometric approach to Cramer's rule: the (generalized) cross product is a vector normal to its arguments with a magnitude proportional to their spanned (hyper-)area. Therefore, the $\mathbf{c}_j$ vectors represent the directed areas of the "base" of each of parallelepipeds from the volumetric view of Cramer's rule. Then taking the inner product of $\mathbf{a}_j$ or $\mathbf{b}$ with $\mathbf{c}_j$ gives the (hyper-)volume of the appropriate parallelepiped. This connection to the (generalized) cross product also provides an explanation for the sign of the determinants, which Orr's presentation neglects.



(a) $\langle c_1, b \rangle$ and $\langle c_1, a_1 \rangle$



(b) $\langle c_2, b \rangle$ and $\langle c_2, a_2 \rangle$



(c) $\langle c_3, b \rangle$ and $\langle c_3, a_3 \rangle$

Figure 0.3: Illustration of the unit vectors $\hat{c}_i$ parallel to the rows of the adjugate of A and their associated inner products for the example problem.

Generalization of projection onto orthogonal vectors

Here we show that the interpretation of (0.7) can be seen as a generalization of writing $\mathbf{b}$ as a linear combination of a set of orthogonal vectors $\mathbf{v}_j$,

$$\mathbf{b} = \sum_{j=1}^n \frac{\langle \mathbf{v}_j, \mathbf{b} \rangle}{\langle \mathbf{v}_j, \mathbf{v}_j \rangle} \mathbf{v}_j. \quad (0.11)$$

In the case of mutually orthogonal columns of A , we can substitute $\mathbf{a}_j$ in (0.7) for $\mathbf{v}_j$. It is clear that the corresponding $\mathbf{c}_j$ must point in the same direction as $\mathbf{v}_j$ because they are both orthogonal to the remaining columns of A . Then $\mathbf{c}_j$ differs from $\mathbf{v}_j$ by a scaling constant $s_j = |\mathbf{v}_j| / |\mathbf{c}_j|$. Multiplying the j th coefficient of (0.7) by $1 = s_j/s_j$ and distributing it into the inner products allows us to substitute $s_j \mathbf{c}_j$ for $\mathbf{v}_j$. Together, these substitutions of (0.7) become (0.11), showing that orthogonal vector expansion is a special case of this interpretation of Cramer's rule.

Application to least squares

The expansion (0.7) shares other properties with an orthogonal expansion. For example, such an expansion can be used to represent the least squares projection onto the subspace spanned by a subset of the columns of A , with the coefficients of the expansion again the ratio of inner products with a generalized cross product. To be consistent with typical projection problems, we'll consider the matrix consisting of m columns of A , $A_m \in \mathbb{R}^{n \times m}$, $m < n$, with A_m full rank, although the result can be extended to relax these conditions. The least squares projection into the space spanned by the columns of A_m is given by

$$\mathbf{p} = A_m (A_m^T A_m)^{-1} A_m^T \mathbf{b}. \quad (0.12)$$

For this discussion, we will use a compact form of the singular value decomposition (SVD), $A_m = U \Sigma V^T$, where the columns of $U \in \mathbb{R}^{n \times m}$ are the orthonormalized eigenvectors associated with the m largest eigenvalues of $A_m A_m^T$, the columns of $V^T \in \mathbb{R}^{m \times m}$ are the orthonormalized eigenvectors of $A_m^T A_m$, and $\Sigma \in \mathbb{R}^{m \times m}$ is a diagonal matrix whose elements are the non-negative square roots of the eigenvalues of $A_m^T A_m$ called the singular values of A_m [6].

Using the SVD, we'll form a pseudo-adjugate as

$$C_m^T = (\det \Sigma) V \Sigma^{-1} U^T, \quad (0.13)$$

which has the same form of the adjugate of (0.8) with $\det \Sigma$ used in place of $\det A_m$, which is not defined, and the pseudo-inverse $A_m^\dagger = V \Sigma^{-1} U^T$ in place of the inverse, which does not exist. Using $\det \Sigma$ in place of $\det A_m$ is reasonable

since the magnitude of the determinant of a square and non-singular matrix is equal to the product of its singular values, which in turn is equal to the determinant of its corresponding Σ matrix.

We also use the SVD to find the factors $A_m^T = V\Sigma U^T$ and $(A_m^T A_m)^{-1} = V^T (\Sigma^{-1})^2 V$. Substituting these into (0.12), carrying out the multiplication of the orthogonal matrices with their inverse, and factoring the scalar $\det \Sigma$ to substitute (0.13), we have

$$\mathbf{p} = A_m \frac{1}{\det \Sigma} C_m^T \mathbf{b},$$

where we have placed the scalar $\frac{1}{\det \Sigma}$ in a convenient order without changing the matrix-vector product. Since $C_m^T A_m = I \det \Sigma$, we can substitute $(C_m^T A_m)^{-1}$ for $\frac{1}{\det \Sigma} I^{-1}$, letting us write the projection as

$$\mathbf{p} = A_m (C_m^T A_m)^{-1} C_m^T \mathbf{b}.$$

In this form, we recognize the vector $\mathbf{x} = (C_m^T A_m)^{-1} C_m^T \mathbf{b}$ is a matrix expression equivalent to (0.6). Then we can interpret the remaining matrix-vector multiplication as

$$\mathbf{p} = \sum_{j=1}^m \frac{\langle \mathbf{c}_j, \mathbf{b} \rangle}{\langle \mathbf{c}_j, \mathbf{a}_j \rangle} \mathbf{a}_j,$$

just as one would write the expansion in terms of orthogonal vectors. In this case, $\mathbf{c}_j$, the j th column of C_m , is the generalized cross product of the $m - 1$ columns of A_m excluding $\mathbf{a}_j$ in the m -dimensional column space of A_m . The vector $\mathbf{c}_j$ depends on all m vectors which define the subspace in which the cross product occurs. So unlike as in the orthogonal case, the coefficient associated with $\mathbf{a}_j$ for the least squares projection depends on the other vectors composing A_m .

We illustrate this projection interpretation using each permutation of two columns of A from Example (0.1) in Figure 0.4. These examples illustrate the property that the rows of the pseudo-adjugate span the same space as the columns of A_m which follows from the position of U in each expression. These examples also illustrate the behavior of the cross product in two dimensions. In each case, the $\mathbf{c}_j$ vectors are perpendicular to the $\mathbf{a}_i$ for $i \neq j$ (while remaining in the plane spanned by $\mathbf{a}_i$ and $\mathbf{a}_j$) as expected. As in the general case, the order of the columns of A_m determines the handedness of A_m 's column space which, together with the skew symmetry sign rule (0.10), determines the direction of the $\mathbf{c}_j$ vector. When applied to two dimensions, each $\mathbf{c}_j$ is rotated from $\mathbf{a}_i$ by 90 degrees in the plane towards $\mathbf{a}_j$. Since the angle between the columns of each A_m for the example problem is less than 90 degrees, each $\mathbf{c}_j$ appears outside of the cone formed by the columns of A_m on the same side as $\mathbf{a}_j$.

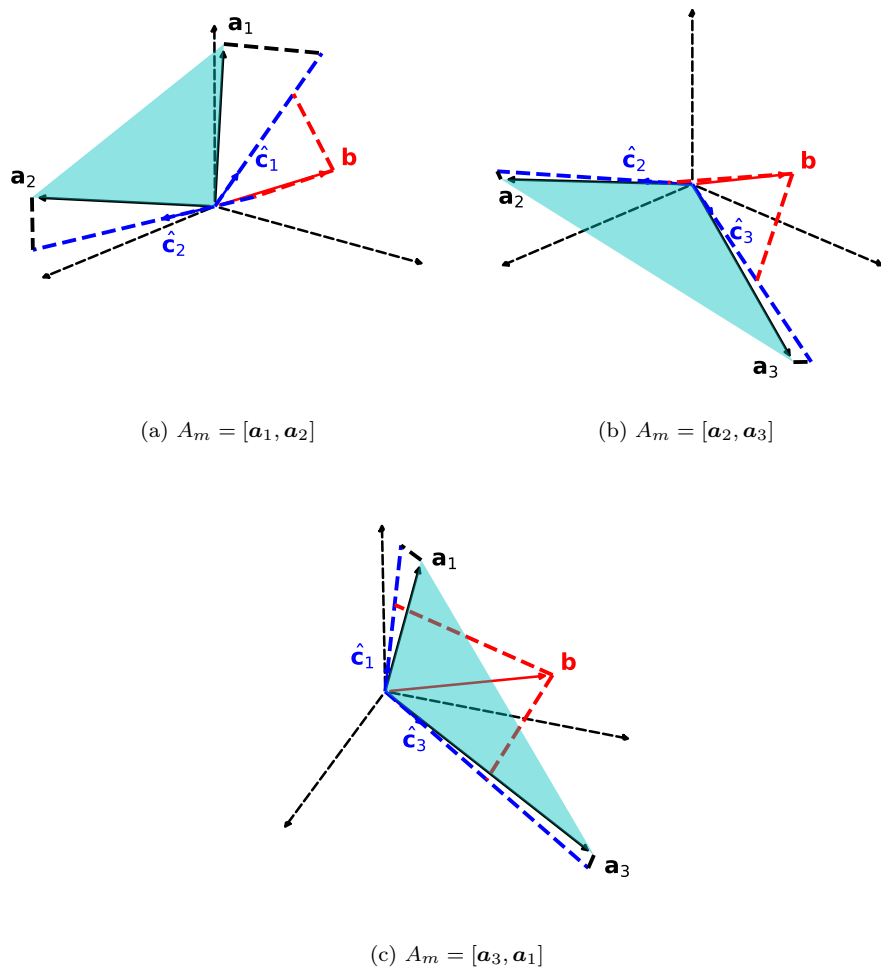


Figure 0.4: Using this interpretation of Cramer's rule to illustrate a projection onto a subspace spanned by any two columns of the example A .

Conclusion

Interpreting Cramer's rule as a generalization of projection onto orthogonal vectors using the adjugate enhances the connections between many elementary linear algebra concepts. Such connections can help make these concepts more cohesive and similar connections were essential to the author's mathematical education as an engineer. Although this article does not provide any immediate practical result, this approach may provide insights into other vector spaces like polynomial spaces which are often treated with non-orthonormal bases.

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